



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-3)

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Jump Statements

- A jump statement is a statement that immediately changes the normal flow of program execution by transferring control from one place to another.
- e.g →
 - break
 - continue
 - goto
 - return

1. Break Statement

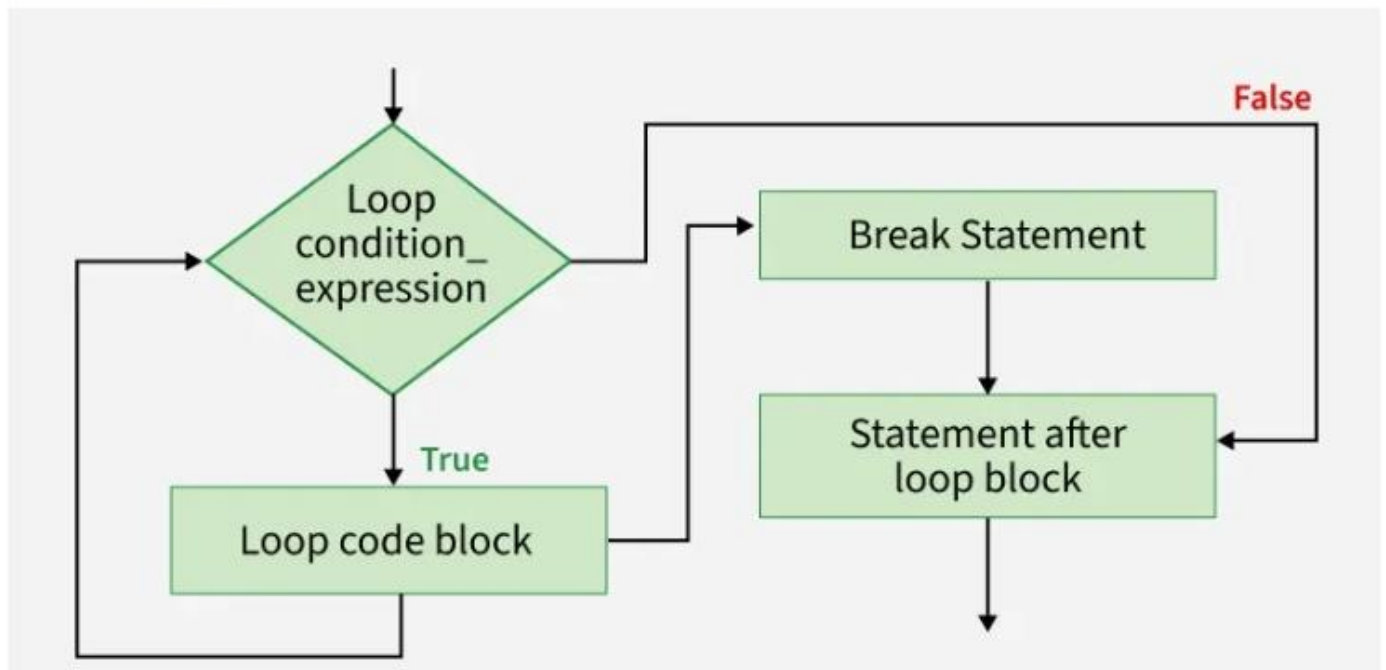
- The break statement in c++ is used to immediately stop the execution of a loop or a switch statement and transfer control to the statement that comes after the loop or switch.
- It is mainly used inside :- for loop, while loop, do-while loop, switch statement.
- **Syntax**

break;

➤ Key Points

- Used with for, while, and do-while loops.
- Also used with switch.
- Terminates the **nearest** loop or switch.
- The remaining iterations are not executed.
- Control moves to the statement after the loop or switch.
- It is commonly used when a required condition is found.
- In nested loops, it terminates only the **innermost loop** in which it occurs.

Flowchart of break Statement



• Example :

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
    int i;
    for (i=1; i<=100;i++)
    {
        if(i==20)
            break;
        cout<< " a = " << i << "\t\t" ;
    }
    getch();
}
```

The screenshot shows the output of the C++ program. The numbers 1 through 19 are printed in a grid format, with 5 columns and 4 rows. The last row contains only 4 numbers (16, 17, 18, 19) because the loop was broken at i=20. The output is as follows:

a = 1	a = 2	a = 3	a = 4	a = 5
a = 6	a = 7	a = 8	a = 9	a = 10
a = 11	a = 12	a = 13	a = 14	a = 15
a = 16	a = 17	a = 18	a = 19	

1. Write a C++ program to print numbers from 1 to 100. Stop the loop when the number 50 is reached using break.
2. Write a C++ program to read integers from the user until the user enters 0. Use break to terminate the loop.
3. Write a C++ program to find the first number between 1 and 100 that is divisible by both 5 and 7. Use break after finding the number.
4. Write a C++ program to print the multiplication table of a given number. Stop printing when the product becomes greater than 100 using break.

5. Write a C++ program to search for a given number in an array of 10 elements. Stop searching as soon as the element is found using break.
6. Write a C++ program to find the first even number entered by the user. Keep accepting input until an even number is entered, then terminate the loop using break.
7. Write a C++ program to check whether a given number is prime. Use break to exit the loop as soon as a factor is found.
8. Write a C++ program to print the first 10 natural numbers, but stop the loop if the current number becomes equal to a user-entered value using break.
9. Write a C++ program using nested loops to print a pattern. Stop the inner loop when the column number becomes 4 using break.
10. Write a C++ program to continuously display a menu (Add, Subtract, Exit). Terminate the program when the user selects the Exit option using break.

2. Continue

- The continue statement is used to skip the remaining statements of the current iteration of a loop and transfer control to the next iteration of that loop.
- Syntax :

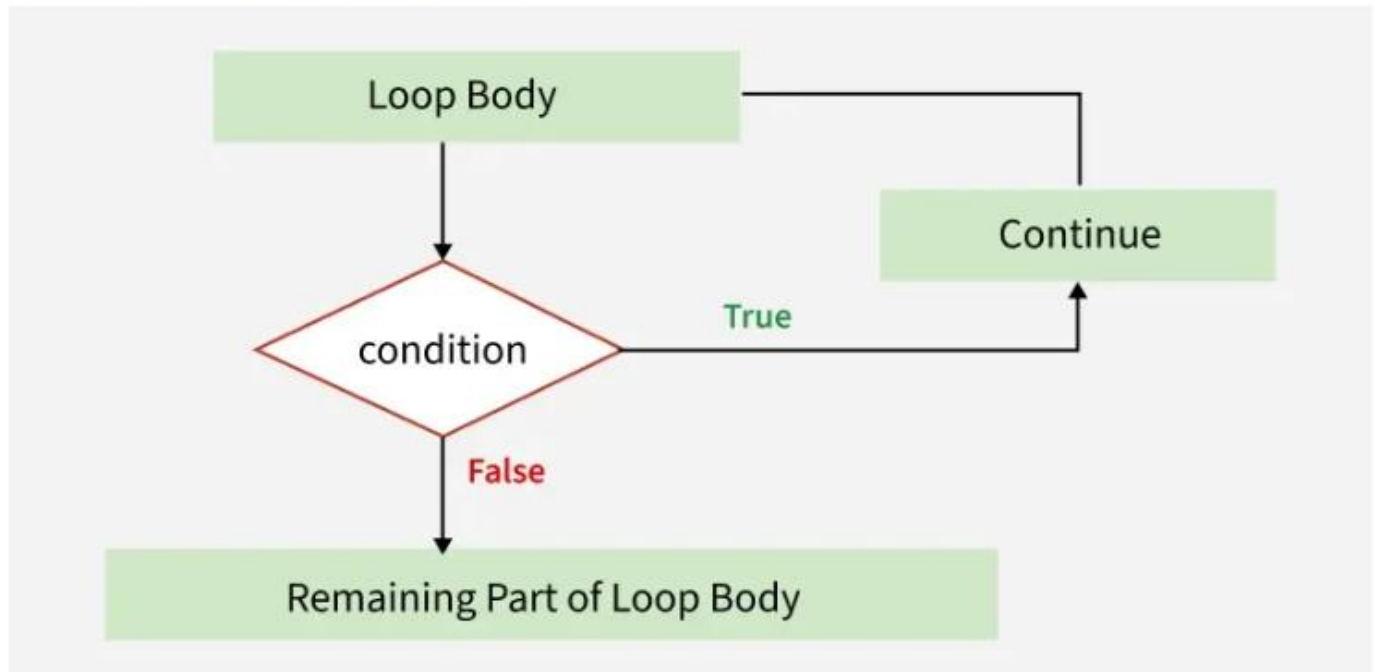
continue;



Key Points

- Used only with loops.
- It does not terminate the loop.
- It skips only the current iteration.
- The next iteration starts after continue.
- Can be used with for, while, and do-while.
- In a for loop, after continue, the update expression is executed.
- In nested loops, it affects the nearest loop.

Flowchart of continue Statement



• Example :

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
    for(int i = 1; i <= 10; i++)
    {
        if(i % 2 == 0)
            continue;

        cout << i << " ";
    }
    getch();
    return 0;
}
```

C:\Users\hira\Desktop\hello.exe

1 3 5 7 9

Questions

1. Predict the output:

```
int main()
{
    int sum = 0;

    for(int i = 1; i <= 10; i++)
    {
        if(i % 3 == 0)
            continue;
```

```
        sum += i;
    }

    cout << sum;
}
```

2. Predict the answer

```
int main()
{
    for(int i = 1; i <= 5; i++)
    {
        for(int j = 1; j <= 5; j++)
        {
            if(j == 3)
                continue;

            cout << j << " ";
        }
        cout << endl;
    }
}
```

3. Predict the answer

```
int main()
{
    int i = 0;

    while(i < 10)
    {
        i++;

        if(i == 4 || i == 7)
            continue;

        cout << i << " ";
    }
}
```

4. Predict the output

```
int main()
{
    int count = 0;

    for(int i = 1; i <= 20; i++)
    {
        if(i % 2 == 0)
            continue;

        if(i % 3 == 0)
            continue;

        count++;
    }

    cout << count;
}
```

5. Predict the answer

```
int main()
{
    int sum = 0;

    for(int i = 1; i <= 10; i++)
    {
        if(i == 5)
            continue;

        if(i % 2 == 0)
            sum += i;
    }

    cout << sum;
}
```

3. goto

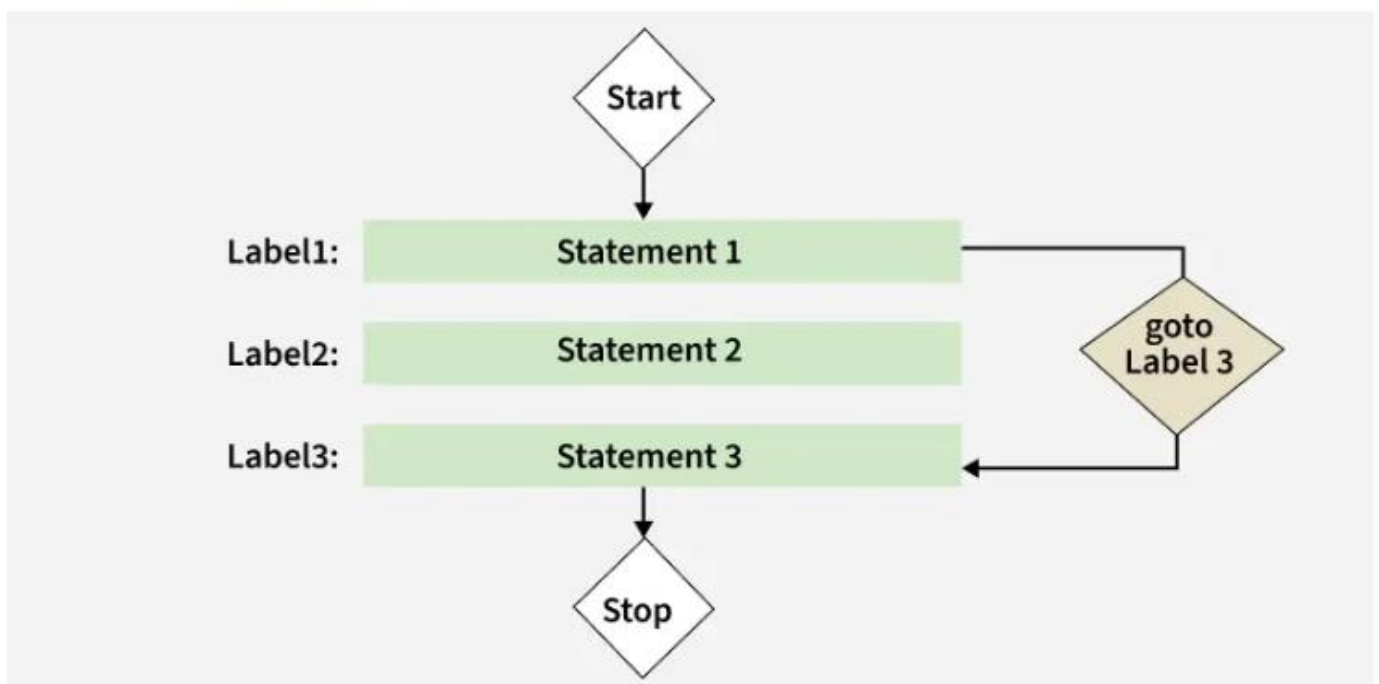
- The goto statement is used to unconditionally transfer the control of program execution to another statement identified by a label within the same function.
- A label is an identifier followed by a colon (:).
- Syntax :
 goto hira;

```
hira:  
    statement;
```

➤ Key Points

- It performs an unconditional jump.
- It transfers control to a labelled statement.
- The label must be within the same function.
- Syntax uses the keyword goto.
- A label is followed by a colon (:).
- It can create loops manually.
- Excessive use of goto can make a program difficult to read and maintain.
- It is generally avoided in structured programming.

Flowchart of goto Statement

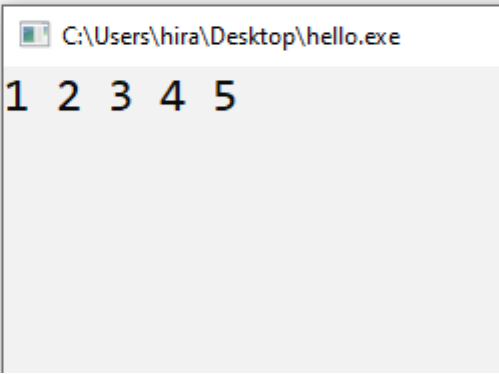


• Example :

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
    int i = 1;

start:
    cout << i << " ";
    i++;

    if(i <= 5)
        goto start;
    getch();
    return 0;
}
```



Questions

1. Write a C++ program using goto to print numbers from 1 to 10.
2. Write a C++ program using goto to find the sum of numbers from 1 to 20.
3. Write a C++ program using goto to print all even numbers between 1 and 50.
4. Write a C++ program using goto to find the factorial of a given number.
5. Write a C++ program using goto to check whether a given number is prime or not.

4. return

- The return statement is used to terminate the execution of the current function and transfer control back to the calling function. It can also return a value from a function to the calling function.

- Syntax :

For Value returning function

return expression;

For a void function

return;

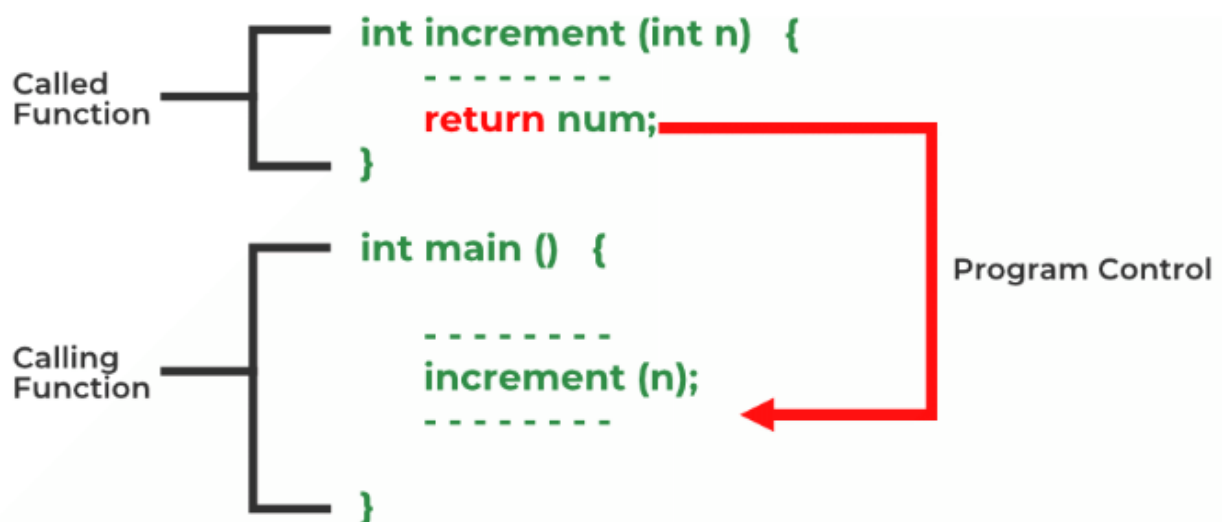


Key Points

- Used inside functions.
- Terminates the current function.
- Can return a value.

- The returned value must be compatible with the function's return type.
- A void function can use return; without a value.
- return 0; in main() traditionally indicates successful program termination.
- When return is executed, statements after it in the same function are not executed.
- A function may contain multiple return statements.

Flowchart of return Statement



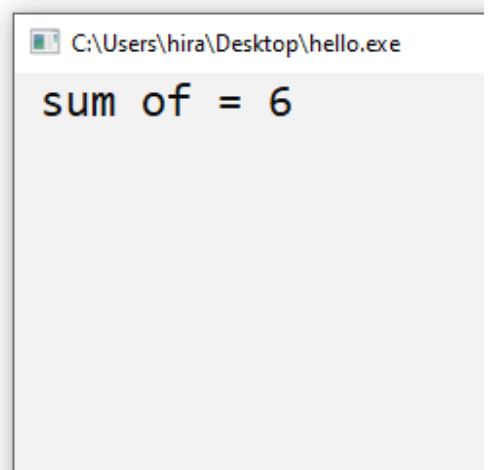
• Example :

```

#include<iostream>
#include<conio.h>
using namespace std;

int add(int a, int b)
{
    return a + b;
}

int main()
{
    int x=1, y=5;
    cout << " sum of = " << add(x,y); // function call
    getch();
    return 0;
}
  
```



❖ Compound statement

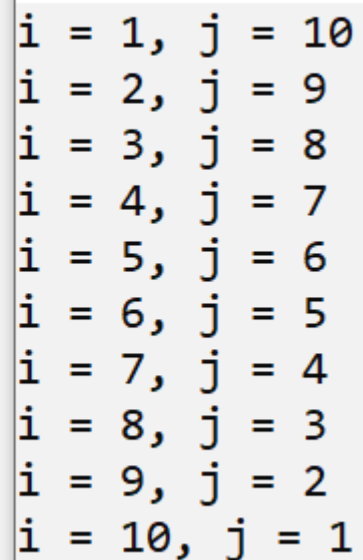
- When we put more than one statement in a block, it is called compound statement.

- Example :

```
#include<iostream>
#include<conio.h>
using namespace std;

int main()
{
    int i,j;
    for (i=1, j=10;i<=10;i=i+1,j=j-1)
    {
        cout << "i = " << i << ", j = " << j << endl;
    }

    getch();
    return 0;
}
```



```
i = 1, j = 10
i = 2, j = 9
i = 3, j = 8
i = 4, j = 7
i = 5, j = 6
i = 6, j = 5
i = 7, j = 4
i = 8, j = 3
i = 9, j = 2
i = 10, j = 1
```

Q1. Write a C++ program using a while loop that repeatedly asks the user to enter a number and displays its multiplication table. The program should continue asking for different numbers until the user chooses to stop.

```
#include<iostream>
using namespace std;
int main()
{
    int n, choice = 1;

    while(choice == 1)
    {
        cout << "Enter a number: ";
        cin >> n;

        for(int i = 1; i <= 10; i++)
        {
            cout << n << " x " << i << " = " << n * i << endl;
        }

        cout << "\nEnter 1 to continue or 0 to stop: ";
    }
}
```

```

        cin >> choice;
    }

    return 0;
}

```

```

using namespace std;
int main()
{
    int n, choice = 1;

    while(choice == 1)
    {
        cout << "Enter a number: ";
        cin >> n;

        for(int i = 1; i <= 10; i++)
        {
            cout << n << " x " << i << " = " << n * i << endl;
        }

        cout << "\nEnter 1 to continue or 0 to stop: ";
        cin >> choice;
    }

    return 0;
}

```

C:\Users\hira\Desktop\hello.exe

Enter a number: 2

```

2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

```

Enter 1 to continue or 0 to stop: 1

Enter a number: 3

```

3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

```

Enter 1 to continue or 0 to stop: 0

Process returned 0 (0x0) execution time : 16.4
Press any key to continue.

=====HIRA KUMAR=====